

SOLVING THE LIGAND-CHANNEL PUZZLE: AN INTEGRATED STRUCTURE-FUNCTION-BINDING PIPELINE.

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ABSTRACT

Ion channels are major transmembrane therapeutic and agricultural targets that translate extracellular chemical signals into detectable electrical currents. Deciphering the molecular mechanisms of their regulation by ligands remains a technical challenge. No single technique is generally sufficient, and it requires a coordinated workflow to bridge the gap between static structures and dynamic functions. Here, we present a combination of interdisciplinary techniques that are complementary. As a structural assay, we use the cryo-EM technique, which provides high-resolution structural snapshots but lacks functional context. To address this issue, we combine functional analysis with the Two-Electrode Voltage Clamp (TEVC) electrophysiological technique with rational site-directed mutagenesis. As a binding assay, we use MicroScale Thermophoresis (MST) to quantify the biophysical features of ligand-protein interactions.

This workflow helps us to decipher key protein-peptide complexes involved in Alzheimer's disease and to uncover new regulatory molecules of insect olfactory receptors for health and crop protection by transforming each individual technical constraint into a solved puzzle.